

Watch-DevKit-HW Fabrication Document

Layer Stack Legend

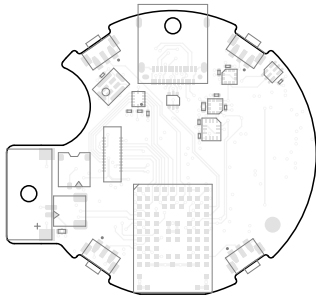
	Material	Layer	Thickness	Dielectric	Type	Gerber
		F.Paste			Paste Mask	
		F.Silkscreen			Legend	GBR
		F.Mask	0,01mm		Solder Mask	GBR
	Copper	F.Cu	0,035mm (1,00oz)		Signal	GBR
	Core		0,196mm	FR4	Dielectric	
	Copper	Inner1	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,43mm	FR4	Dielectric	
	Copper	Inner2	0,035mm (1,00oz)		Plane	GBR
	Core		0,196mm	FR4	Dielectric	
	Copper	B.Cu	0,035mm (1,00oz)		Signal	GBR
	B.Mask		0,01mm		Solder Mask	GBR
	B.Silkscreen				Legend	GBR
	B.Paste				Paste Mask	

Total thickness: 0,982mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [%]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
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Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- SURFACE FINISH: HAL LEAD-FREE
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR WHITE.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING BLACK NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 150 C OR EQUIVALENT.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

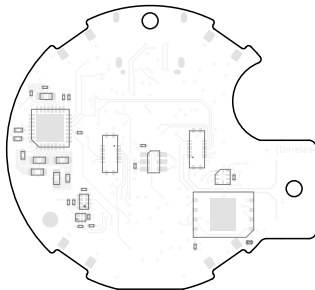
BOARD SIZE 40,972 × 37,500 mm
BOARD THICKNESS 0,982 mm
TRACE WIDTH 0,150 mm
TRACE TO TRACE 0,125 mm
MIN. HOLE (PTH) 0,250 mm
MIN. HOLE (NPTH) 0,650 mm
ANNULAR RING 0,100 mm
COPPER TO HOLE 0,125 mm
COPPER TO EDGE 0,125 mm
HOLE TO HOLE 0,250 mm

	Comments: ZSWatch	Company: ZSWatch		Variant: CHECKED	Git Hash: f34db5b
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb		Designer: Daniel Kampert	Date: 2025-07-13
	Sheet Path:		Revision: + (Unreleased)	Size: A3	Sheet: 1 of 10

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Bottom Fabrication (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
	ZSWatch	ZSWatch		CHECKED	f34db5b
		Board Name:		Project Name:	
		Watch-DevKit-HW		ZSWatch Watch-DevKit	
Sheet Title:	File Name:	Designer:	Date:	Revision:	
Bottom Fabrication (Scale 1:1)	ZSWatch-Watch.kicad_pcb	Daniel Kampert	2025-07-13	+ (Unreleased)	
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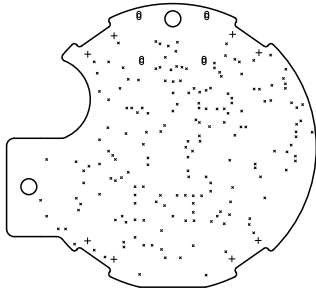
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Watch-DevKit-HW Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	182	0,25mm (9,84mils)	PTH	Round	F,Cu - B,Cu	Via
O	4	0,50mm (19,69mils)	PTH	Slot	F,Cu - B,Cu	Pad
+	8	0,80mm (31,50mils)	PTH	Round	F,Cu - B,Cu	Pad
	Total 194					



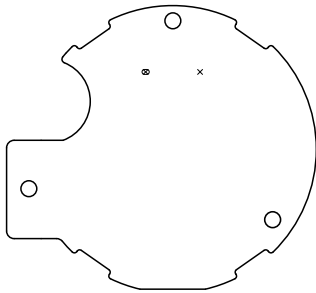
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		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10

Watch-DevKit-HW Fabrication Document

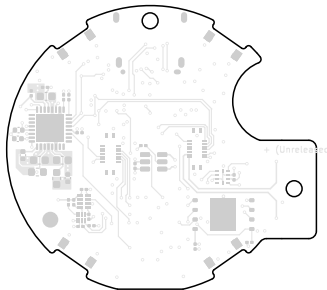
Drill Drawing L1 - L4 (Scale 1:1)

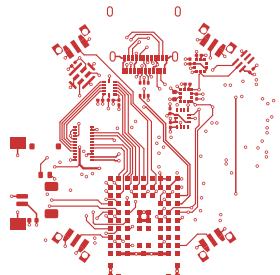
Drill Table

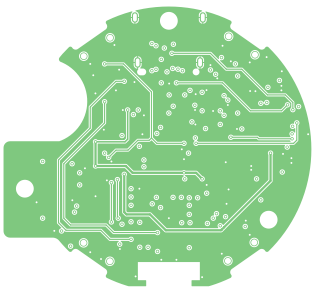
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	1 + 1	0,65mm (25,59mils)	NPTH	Round + Slot	F,Cu - B,Cu	Mechanical
O	1	2,10mm (82,68mils)	NPTH	Round	F,Cu - B,Cu	Mechanical
	Total 3					



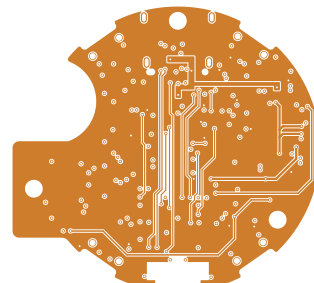
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		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 4 of 10

Watch-DevKit-HW Fabrication Document							
Bottom Test Points (Scale 1:1)							
							
		Comments:	Company:		Variant:	Git Hash:	
		ZSWatch	ZSWatch		CHECKED	f34db5b	
			Board Name:		Project Name:		
			Watch-DevKit-HW		ZSWatch Watch-DevKit		
		Sheet Title:	File Name:	Designer:	Date:	Revision:	
		Bottom Test Points (Scale 1:1)	ZSWatch-Watch.kicad_pcb	Daniel Kampert	2025-07-13	+ (Unreleased)	
		Sheet Path:			Reviewer:	Size:	Sheet:
						A3	6 of 10

Watch-DevKit-HW Fabrication Document												
F.Cu (Scale 1:1)												
												
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							Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit			
					Sheet Title: F.Cu (Scale 1:1)		File Name: ZSWatch-Watch.kicad_pcb		Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)	
					Sheet Path:			Reviewer:		Size: A3	Sheet: 7 of 10	

	1	2	3	4	5	6	7	8	
A	Watch-DevKit-HW Fabrication Document Inner1 (Scale 1:1) 								A
B									B
C									C
D									D
E									E
F	1	2	3	4	5	6	7	8	F

Comments: ZSWatch	Company: ZSWatch		Variant: CHECKED		Git Hash: f34db5b	
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit			
Sheet Title: Inner1 (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb		Designer: Daniel Kampert		Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:			Reviewer:		Size: A3	Sheet: 8 of 10

Watch-DevKit-HW Fabrication Document							
Inner2 (Scale 1:1)							
							

Watch-DevKit-HW Fabrication Document							
B.Cu (Scale 1:1)							
